

MUHAMMAD ARGYA VITYASY

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EDUCATION

Universitas Gadjah Mada

Yogyakarta, Indonesia

Bachelor of Computer Science | Cumulative GPA: 3.85/4.00 (113/144 credits)

August 2023 – Present

- **Relevant Coursework:** Algorithms and Data Structures, Scalable Software Engineering, Software Engineering Methods, Cloud Computing, Computer Networks, Cyber Security

WORK EXPERIENCE

Software Engineer Intern | GDP Labs

February 2026 – June 2026

Enterprise AI automation platform building multi-agent LLM systems for HR, payroll, and project management workflows.

- Built LLM-powered agents in a multi-agent system automating enterprise HR and project management workflows, reducing manual intervention across 3+ workflow types.
- Designed a compliance monitoring pipeline that auto-generates audit summaries, eliminating manual report tracking across organizational workflows.
- Architected a reusable, multi-tenant IAM SDK to standardize authentication across enterprise projects, eliminating redundant setup while preventing partial-state failures via atomic multi-provider transactions.
- Authored Playwright end-to-end test suites to catch regressions and guarantee reliable multi-agent tool execution before deployment to production.

Backend Engineer | PT Satria Duta Raya

October 2025 – January 2026

B2B procurement platform enabling businesses to discover ranked products through an asynchronous recommendation pipeline.

- Architected a Go/Gin backend enforcing 6 strict domain boundaries to prevent tight coupling and enable independent feature scaling across a B2B procurement platform.
- Built an event-driven product ranking pipeline on Redis Streams with idempotent consumers and Dead Letter Queues, decoupling heavy computation from request paths and containing failure blast radius.
- Shipped full-stack observability using Prometheus and Grafana to proactively identify performance bottlenecks and drastically reduce Mean Time to Resolution (MTTR) for background worker failures.

Backend Engineer | Jogja International Model United Nation 2025

April 2025 – October 2025

Annual academic simulation of the United Nations bringing together students worldwide for diplomatic debates.

- Maintained 99.9% uptime during 500+ concurrent registration spikes by optimizing database connection pooling and leveraging Docker containerization with Go and PostgreSQL.
- Cut manual staff workload by 40% by building a role-based admin dashboard that automated user verification and approval workflows for 150+ staff actions.

Backend Engineer | Organisasi Mahasiswa Ahli Teknologi Informasi UGM

March 2024 – July 2025

Student-led IT organization comprising 8 specialized divisions and 80+ members managing real-world software projects.

- Built a reusable Go/Node.js/Docker backend framework deployed across 4 client projects, cutting development and deployment time by 60%.
- Handled 3,000+ user registrations and high-concurrency workloads with 99.9% uptime across all deployed projects.
- Led weekly technical discussions for 15 engineers, facilitating deep-dives into distributed systems, Kubernetes, and secure authentication to drive team-wide technical growth.

PROJECTS

National University Entrance Exam Mock Platform | [GitHub](#)

- Served 200+ concurrent users at sub-50ms latency across a 4-service Go/Docker microservices architecture.
- Processed 150+ timed exam sessions from a 1,000-question bank with <5% error rate via token-based auth and a custom exam engine exposing 20+ REST APIs.
- Designed a migration plan to Kafka-based async processing and Kubernetes orchestration for future scalability.

ADDITIONAL SKILLS & INTERESTS

Languages & Frameworks: Go, Python, TypeScript, Gin-Gonic, Express, Pydantic

Data & Infrastructure: PostgreSQL, MongoDB, Redis, Kafka, Docker, Colima, Terraform, Ansible, AWS S3, MinIO

Observability & Testing: Prometheus, Grafana, Playwright

Interests: LLM-powered automation, distributed systems, cloud computing, DevSecOps, system design & scalability